

▼ Problem 03

These data were taken from Kelly (1984) and consist of simultaneous pairs of measurements of serum kanamycin levels in blood samples drawn from twenty premature babies. One of the measurements was obtained by a Heelstick method (x) and the other by using an umbilical Catheter (y). Since there is a measurement error in both methods, the measurement error model seems to be appropriated for describing these data.

Estimate the regression parameters by the instrumental variable methods proposed by Wald, Bartlett and Durbin. Also, compare these three methods.

Y	X
67.50	80.09
75.44	72.57
109.7	112.14
105.44	121.57
109.4	125.60
85.83	131.48
76.7	131.54
129.42	145.60
104.24	168.56
125.83	171.48
153.99	203.54
152.92	222.85
160.03	230.93
176.33	232.99
174.53	261.18

```
# Data
Y <- c(67.50,75.44,109.7,105.44,109.4,85.83,76.7,129.42,104.24,125.83,153.99,152.92,160.03,176.33,174.53)
X <- c(80.09,72.57,112.14,121.57,125.60,131.48,131.54,145.60,168.56,171.48,203.54,222.85,230.93,232.99,261.18)

# OLS and MSE
ols <- lm(Y ~ X)
mse <- summary(ols)$sigma^2

# Sort data
data <- data.frame(Y, X)[order(X),]
x <- data$X
y <- data$Y

# Function for IV estimation
iv_est <- function(x, y, z) {
  x.mat <- cbind(1, x)
  z.mat <- cbind(1, z)
  b.hat <- solve(t(z.mat) %*% x.mat) %*% (t(z.mat) %*% y)
  var.mat <- mse * (solve(t(z.mat) %*% x.mat) %*% (t(z.mat) %*% z.mat) %*% solve(t(x.mat) %*% z.mat))
  data.frame(a = b.hat[1], b = b.hat[2], var.a = var.mat[1,1], var.b = var.mat[2,2])
}

### Wald Method ###
z1 <- ifelse(x <= median(x), -1, 1)
wald <- iv_est(x, y, z1)

### Bartlett Method ###
n1 <- round(length(x)/3)
n2 <- 2*n1 + 1
x1 <- c(x[1:n1], x[n2:length(x)])
y1 <- c(y[1:n1], y[n2:length(x)])
z2 <- ifelse(x1 <= median(x1), -1, 1)
bartlett <- iv_est(x1, y1, z2)

### Durbin Method ###
z3 <- 1:length(x)
durbin <- iv_est(x, y, z3)

# Show results
wald
bartlett
durbin
```

```

      A data.frame: 1 × 4
      a      b    var.a    var.b
  <dbl>  <dbl>  <dbl>  <dbl>
1 30.62147 0.5588354 157.907 0.0055897

      A data.frame: 1 × 4
      a      b    var.a    var.b
  <dbl>  <dbl>  <dbl>  <dbl>
1 37.40602 0.5477858 155.6462 0.004900569

      A data.frame: 1 × 4
      a      b    var.a    var.b
  <dbl>  <dbl>  <dbl>  <dbl>
1 29.51776 0.5656989 126.1139 0.004360233

```

✓ Problem 04

- Fit a linear regression model of Y on X.
- Draw a scatter plot of X and Y to identify unusual observations.
- Identify influential observations, high leverage points, and outliers (if any) from the above data.
- Repeat analysis after changing the 9th Y value from 79.24 to 65.24.
- Repeat analysis with additional modifications: 9th Y from 79.24 to 65.24, 22nd Y from 52.32 to 35.32, and 22nd X from 810 to 610.
- Comment on your findings

Y	X
16.68	560
11.50	220
12.03	340
14.88	80
13.75	150
18.11	330
8.00	110
17.83	210
79.24	1460
21.50	605
40.33	688
21.00	215
13.50	255
19.75	462
24.00	448
29.00	776
15.35	200
19.00	132
9.50	36
35.10	770
17.90	140
52.32	810
18.75	450
19.83	635
10.75	150

```

# Original data
y <- c(16.68,11.50,12.03,14.88,13.75,18.11,8.00,17.83,79.24,21.50,40.33,21.00,13.50,
      19.75,24.00,29.00,15.35,19.00,9.50,35.10,17.90,52.32,18.75,19.83,10.75)
x <- c(560,220,340,80,150,330,110,210,1460,605,688,215,255,462,448,776,
      200,132,36,770,140,810,450,635,150)

analyze <- function(y, x, title) {

```

```

model <- lm(y ~ x)
plot(x, y, pch=19, col="blue", main=title)
abline(model, col="red", lwd=2)

h <- hatvalues(model)           # leverage
r <- rstudent(model)           # studentized residuals
c <- cooks.distance(model)     # Cook's distance
n <- length(y); p <- 2
cut <- 2 * p / n               # leverage cutoff

cat("\n----", title, "----\n")
cat("Model")
print(model)
cat("High leverage points:", which(h > cut), "\n")
cat("Outliers (|r|>3):", which(abs(r) > 3), "\n")
cat("Influential (CookD>1):", which(c > 1), "\n")
cat("R-squared:", summary(model)$r.squared, "\n")
}

```

Using Function

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum (x_i - \bar{x})^2}$$

$$\text{Standard residuals, } d_i = \frac{e_i}{\sqrt{MS_{Res}}}$$

$$MS_{Res} = \frac{\sum e_i^2}{n-p}$$

$$\text{Studentized residuals: } r_i = \frac{|e_i|}{\sqrt{MS_{Res}(1-h_{ii})}}$$

$$\text{Cook's distance: } CD_i = \frac{h_{ii}}{p(1-h_{ii})} r_i^2$$

```

# Original data
y <- c(16.68,11.50,12.03,14.88,13.75,18.11,8.00,17.83,79.24,21.50,40.33,21.00,13.50,
      19.75,24.00,29.00,15.35,19.00,9.50,35.10,17.90,52.32,18.75,19.83,10.75)
x <- c(560,220,340,80,150,330,110,210,1460,605,688,215,255,462,448,776,
      200,132,36,770,140,810,450,635,150)

analyze_2 <- function(y, x, title) {
  model <- lm(y ~ x)
  plot(x, y, pch=19, col="blue", main=title)
  abline(model, col="red", lwd=2)

  # High leverage point
  d = (x - mean(x))^2 / sum((x - mean(x))^2)
  n = length(x)
  h = (1/n) + d

  # Outlier
  rs = model$residuals
  msr = sum(rs^2) / (n-p)
  di = abs(rs) / sqrt(msr) # Standardized residuals
  ri = abs(rs) / sqrt(msr * (1 - h)) # studentized residuals

  # Influntial Observation
  cook_dist = (h * ri^2) / (p*(1-h))

  # Cut of point
  p = 2
  cut = 2*(p/n) # Twice of the mean rule

  cat("\n----", title, "----\n")
  cat("Model")
  print(model)
  cat("High leverage points:", which(h > cut), "\n")
  cat("Outliers (|r|>3):", which(ri > 3), "\n")
  cat("Outliers (|d|>3):", which(di > 3), "\n")
  cat("Influential (CookD>1):", which(cook_dist > 1), "\n")
  cat("R-squared:", summary(model)$r.squared, "\n")
}

```

```
analyze_2(y, x, "Original Data")
```

---- Original Data ----

Model

Call:

lm(formula = y ~ x)

Coefficients:

(Intercept) x
4.96116 0.04257

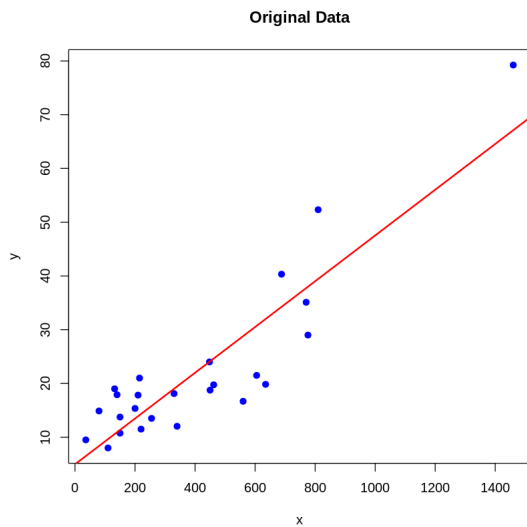
High leverage points: 9

Outliers ($|r| > 3$):

Outliers ($|d| > 3$):

Influential (CookD > 1): 9

R-squared: 0.7950755



1 Original

analyze(y, x, "Original Data")

---- Original Data ----

Model

Call:

lm(formula = y ~ x)

Coefficients:

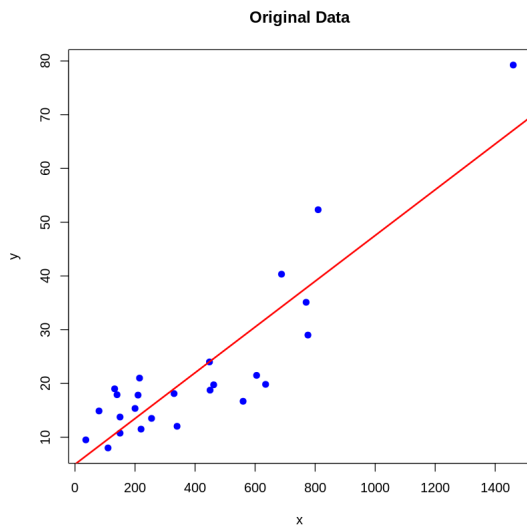
(Intercept) x
4.96116 0.04257

High leverage points: 9

Outliers ($|r| > 3$):

Influential (CookD > 1): 9

R-squared: 0.7950755



```
# 2 Change 9th Y
y2 <- y; y2[9] <- 65.24
analyze(y2, x, "After changing 9th Y = 65.24")
```

---- After changing 9th Y = 65.24 ----

Model

Call:

```
lm(formula = y ~ x)
```

Coefficients:

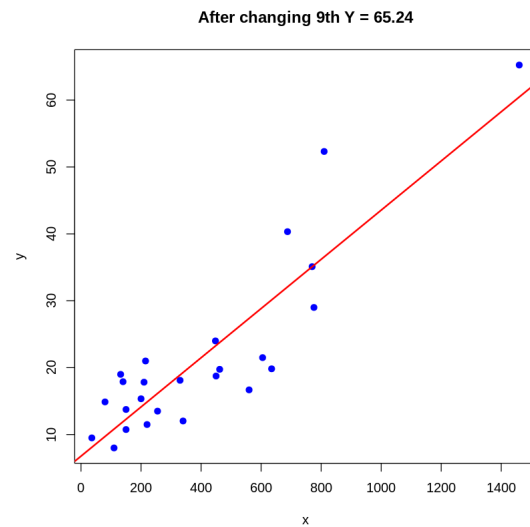
(Intercept)	x
6.77338	0.03677

High leverage points: 9

Outliers ($|r| > 3$): 22

Influential (CookD>1):

R-squared: 0.783432



```
# 3 Change 9th Y, 22nd Y, and 22nd X
y3 <- y; x3 <- x
y3[c(9,22)] <- c(65.24, 35.32)
x3[22] <- 610
analyze(y3, x3, "After changing 9th Y, 22nd Y, and 22nd X")
```

---- After changing 9th Y, 22nd Y, and 22nd X ----

Model

Call:

```
lm(formula = y ~ x)
```

Coefficients:

```
(Intercept)      x
  7.24545      0.03464
```

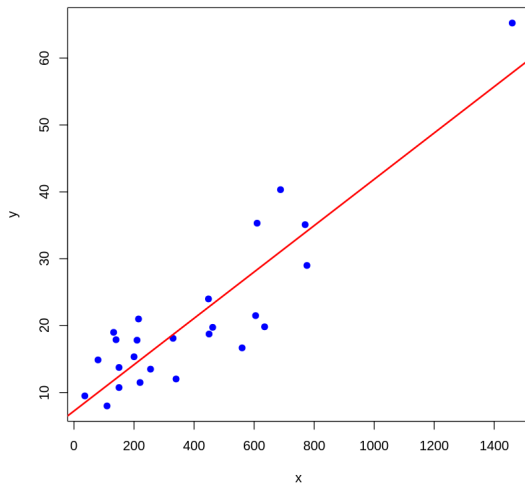
High leverage points: 9

Outliers ($|r| > 3$):

Influential (CookD>1): 9

R-squared: 0.8003564

After changing 9th Y, 22nd Y, and 22nd X



Problem 05

Find the least square and median square regression from the following dataset where fires (y) is dependent variable and years (x) is independent variable.

years = 1976, 1977, 1978, 1979, 1980

fires = 16694, 12271, 12904, 14036, 13874

```
# Data
years <- 1976:1980
fires <- c(16694, 12271, 12904, 14036, 13874)

# (i) Scatter plot
plot(years, fires, col="red", pch=19, main="Scatter Plot of Fires vs Years")

# (ii) Least Squares (LS) regression
ls_model <- lm(fires ~ years)
abline(ls_model, col="blue", lwd=2)
summary(ls_model)

# (iii) Prepare combinations of 3 points
comb <- combn(1:5, 3)
e <- c()

# (iv) Median of squared errors for each combination
for (i in 1:ncol(comb)) {
  idx <- comb[, i]
  x <- years[idx]
  y <- fires[idx]
  model <- lm(y ~ x)
  pred <- predict(model, data.frame(x = years))
  err <- fires - pred
  e[i] <- median(err^2)
}

# (v) Find smallest median error and best combination
```

```

best <- which.min(e)
best_idx <- comb[, best]

cat("\nBest combination (for LMS):", best_idx, "\n")

# (vi) LMS regression using best 3 points
x_best <- years[best_idx]
y_best <- fires[best_idx]
lms_model <- lm(y_best ~ x_best)
summary(lms_model)

# (vii) Plot both lines
abline(lms_model, col="darkgreen", lwd=2, lty=2)
legend("topleft",
      legend=c("Data", "LS line", "LMS line"),
      col=c("red", "blue", "darkgreen"),
      pch=c(19, NA, NA),
      lty=c(NA, 1, 2))

```

Call:

```
lm(formula = fires ~ years)
```

Residuals:

```

      1      2      3      4      5
1963.2 -2072.3 -1051.8  467.7  693.2

```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	780430.8	1139377.2	0.685	0.543
years	-387.5	576.0	-0.673	0.549

Residual standard error: 1822 on 3 degrees of freedom

Multiple R-squared: 0.1311, Adjusted R-squared: -0.1586

F-statistic: 0.4525 on 1 and 3 DF, p-value: 0.5493

Best combination (for LMS): 2 3 5

Call:

```
lm(formula = y_best ~ x_best)
```

Residuals:

```

      1      2      3
-42.29  63.43 -21.14

```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.030e+06	7.245e+04	-14.22	0.0447 *
x_best	5.273e+02	3.662e+01	14.40	0.0441 *

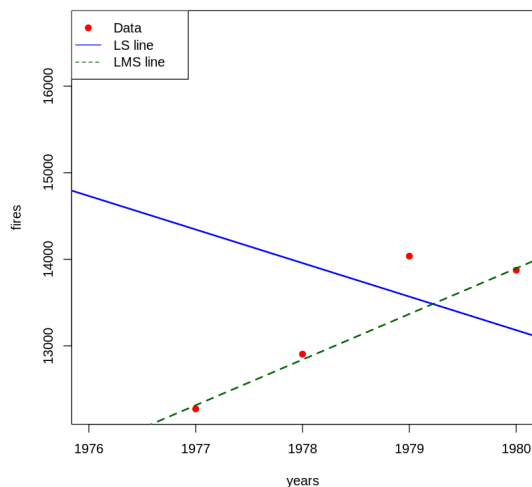
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 79.11 on 1 degrees of freedom

Multiple R-squared: 0.9952, Adjusted R-squared: 0.9904

F-statistic: 207.3 on 1 and 1 DF, p-value: 0.04414

Scatter Plot of Fires vs Years



Problem 06

Fit the LS and LTS (20% trimming) regression from the following dataset, where cost is dependent variable and weight is the independent variable

```
cost <- c(2449, 2248, 3545, 789, 1619, 2079, 918, 1231, 3641, 4314, 2628, 3989,
2308, 376, 5428, 2786, 2497, 5551, 5208, 5438)
weight <-c(90, 87, 38, 28, 28, 23, 21, 17, 27, 39, 34, 46, 80, 14, 48, 38, 73, 40, 44,
41)
```

```
myData = data.frame(cost, weight)
```

```
ls_model = lm(cost ~ weight, data = myData)
```

```
myData$residuals = abs(ls_model$residuals)
```

```
LTS_step = function(data, alpha = 0.2, p =1) {
  n = nrow(data)
  h = round((1 - alpha)*n + (p+1)/2)

  # sort by smallest absolute residuals
  sort_data = data[order(data$residuals), ]

  # keep 1st h data
  trimmed_data = sort_data[1:h, ]

  # Fit new model
  model = lm(cost ~ weight, data = trimmed_data)

  trimmed_data$residuals = abs(model$residuals)
  list(model = model, data = trimmed_data)
}
```

```
# Iteratively apply LTS trimming steps
step1 = LTS_step(myData)
step2 = LTS_step(step1$data)
step3 = LTS_step(step2$data)
step4 = LTS_step(step3$data)
step5 = LTS_step(step4$data)
step6 = LTS_step(step5$data)
```

```
# Final Model
model_ls = ls_model
model_lts1 = step1$model
model_lts2 = step2$model
model_lts3 = step3$model
model_lts4 = step4$model
model_lts5 = step5$model
model_lts6 = step6$model
```

```
# Plot models
plot(weight, cost, main = "All model graph")
abline(model_ls, col=1)
abline(model_lts1, col=2)
abline(model_lts2, col=3)
abline(model_lts3, col=4)
abline(model_lts4, col=5)
abline(model_lts5, col=6)
abline(model_lts6, col=7)

legend("topright", c("LS", paste0("LTS", 1:6)), col = 1:7, lty = 1)
```


All model graph

